

A family of counterexamples and an exact signless-Laplacian resolvent formula

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Status. Likely valid counterexample to Conjecture 8.1 as printed, and likely valid full answer to Problem 8.2 of Hillar–Lin–Wibisono [1]. The stronger motivating version of the conjecture is not settled.

Source and exact questions

C. Hillar, S. Lin, and A. Wibisono, “Inverses of symmetric, diagonally dominant positive matrices and applications,” arXiv:1203.6812, Section 8 [1]. The paper defines $A \geq B$ to mean that $A - B$ is entrywise nonnegative, reserving $A \succeq B$ for Loewner order. With $S(\alpha, m) = \alpha I + m \mathbf{1}\mathbf{1}^\top$, Conjecture 8.1 says that every symmetric diagonally dominant $0 < J \leq S(\alpha, m)$ obeys

$$\|J^{-1}\|_\infty \geq \|S(\alpha, m)^{-1}\|_\infty = \frac{\alpha + 2m(n-1)}{\alpha(\alpha + mn)}, \quad (1)$$

with equality only for $J = S$. Problem 8.2 asks for exact combinatorial formulas for all coefficients of the rational functions $(S + tP)_{ij}^{-1}$ when P is a graph signless Laplacian.

8 Open Problems

As an analogue to Theorem [6.1](#) we also conjecture a tight lower bound for symmetric diagonally dominant matrices $J > 0$ whose off-diagonal entries and diagonal dominances are bounded above. Observe that when $J_{ij} \leq m$ and $\Delta_i(J) \leq \delta$ for all $i \neq j$, then $J \leq (m(n-2) + \delta)I_n + m\mathbf{1}_n\mathbf{1}_n^\top$.

Conjecture 8.1. *Let $n \geq 3$ and let $S(\alpha, m) = \alpha I_n + m\mathbf{1}_n\mathbf{1}_n^\top$. For all $n \times n$ symmetric diagonally dominant matrices $0 < J \leq S(\alpha, m)$, we have*

$$\|J^{-1}\|_\infty \geq \|S(\alpha, m)^{-1}\|_\infty = \frac{\alpha + 2m(n-1)}{\alpha(\alpha + mn)}.$$

Moreover, equality is achieved if and only if $J = S(\alpha, m)$.

In working towards the proof of this conjecture, the following problem may be useful.

Problem 8.2. *Given a signless Laplacian P of a graph G , give an exact combinatorial formula for the entries of $(S + tP)^{-1}$ for any $t > 0$. More precisely, since each entry of $(S + tP)^{-1}$ is a rational function of t , derive a formula for the coefficients of this rational function in terms of the combinatorics of the graph G .*

We also conjecture that a dependence on the largest entry can be removed in Theorem [1.5](#) (b).

Conjecture 8.3. *For a positive, diagonally balanced symmetric J , we have the bound:*

$$\frac{\det(J)}{\prod_{i=1}^n J_{ii}} \leq \frac{\det(S)}{(n-1)^n} = 2 \left(1 - \frac{1}{n-1}\right)^{n-1} \rightarrow \frac{2}{e}.$$

Proof intuition

The printed conjecture has forgotten information present in its motivating paragraph. Entrywise domination allows one to keep every diagonal equal to the diagonal of S while making all off-diagonal entries tiny. The resulting matrix is nearly scalar, so its inverse norm is much smaller than that of S . This produces counterexamples for every parameter choice.

For the graph problem, factor

$$\alpha I + \ell \mathbf{1}\mathbf{1}^\top + tBB^\top = XX^\top$$

using weighted coordinate, all-ones, and unsigned-incidence columns. Cauchy–Binet expands the determinant. Applying the same identity to the complementary minors in the cofactor formula expands every inverse-entry numerator. Counting chosen edge columns reads off every power of t .

A universal counterexample family

Theorem 1. *For every $n \geq 3$ and every $\alpha, m > 0$, Conjecture 8.1 as printed is false. More precisely, for all sufficiently small $\varepsilon > 0$,*

$$J_\varepsilon = (\alpha + m - \varepsilon)I + \varepsilon \mathbf{1}\mathbf{1}^\top \quad (2)$$

is positive entrywise, symmetric, diagonally dominant, and entrywise at most $S(\alpha, m)$, but $\|J_\varepsilon^{-1}\|_\infty < \|S(\alpha, m)^{-1}\|_\infty$.

Proof. The diagonal of J_ε is $\alpha + m$, equal to that of S , and its off-diagonal entries are $\varepsilon \leq m$. It is diagonally dominant provided

$$\alpha + m \geq (n - 1)\varepsilon. \quad (3)$$

Thus every sufficiently small positive ε satisfies all printed hypotheses. The inverse of $aI + b\mathbf{1}\mathbf{1}^\top$ is

$$\frac{1}{a}I - \frac{b}{a(a + nb)}\mathbf{1}\mathbf{1}^\top.$$

Its diagonal inverse entries are positive and its off-diagonal entries are negative. Taking $a = \alpha + m - \varepsilon$ and $b = \varepsilon$ therefore gives

$$\|J_\varepsilon^{-1}\|_\infty = \frac{\alpha + m + (2n - 3)\varepsilon}{(\alpha + m - \varepsilon)(\alpha + m + (n - 1)\varepsilon)}. \quad (4)$$

The right side tends to $1/(\alpha + m)$ as $\varepsilon \downarrow 0$. On the other hand, the proposed lower-bound value in (1) satisfies

$$\|S^{-1}\|_\infty - \frac{1}{\alpha + m} = \frac{m(n - 1)(\alpha + 2m)}{\alpha(\alpha + mn)(\alpha + m)} > 0. \quad (5)$$

Continuity of (4), together with (3), completes the proof. $\square$

For the smallest displayed instance, take $n = 3$ and $\alpha = m = 1$:

$$S = I + \mathbf{1}\mathbf{1}^\top, \quad J = \frac{3}{2}I + \frac{1}{2}\mathbf{1}\mathbf{1}^\top.$$

Then $J \leq S$ entrywise and is diagonally dominant, while

$$S^{-1} = I - \frac{1}{4}\mathbf{1}\mathbf{1}^\top, \quad J^{-1} = \frac{2}{3}I - \frac{1}{9}\mathbf{1}\mathbf{1}^\top, \quad \|S^{-1}\|_\infty = \frac{5}{4} > \frac{7}{9} = \|J^{-1}\|_\infty.$$

Why this does not settle the natural repair

Immediately before Conjecture 8.1, the source assumes separate upper bounds $J_{ij} \leq m$ and $\Delta_i(J) \leq \delta$ and observes that they imply $J \leq S(\alpha, m)$ for $\alpha = m(n - 2) + \delta$. The converse is false. For (2),

$$\Delta_i(J_\varepsilon) - \delta = (n - 1)(m - \varepsilon) > 0 \quad (0 < \varepsilon < m). \quad (6)$$

The example therefore exploits precisely the missing dominance cap. The version retaining both motivating bounds remains open.

An exact answer to the graph resolvent problem

Let $G = (V, E)$ be a loopless graph on $V = \{1, \dots, n\}$, let B be its unsigned incidence matrix, and let $P = BB^\top$. We write

$$M(t) = \alpha I + \ell \mathbf{1}\mathbf{1}^\top + tP, \quad \alpha, \ell, t > 0. \quad (7)$$

Form the ground set $\Omega = V \sqcup \{\star\} \sqcup E$. Associate columns and weights

q	a_q	w_q
$v \in V$	e_v	α
$\star$	$\mathbf{1}$	ℓ
$e = \{u, v\} \in E$	$e_u + e_v$	t .

(8)

Fix an order on Ω . For $F \subseteq \Omega$, let A_F contain its columns in that order and let $w(F) = \prod_{q \in F} w_q$. For $|F| = n - 1$ set

$$\Delta_i(F) = \det[e_i \ A_F], \quad (9)$$

where e_i is the first column.

Theorem 2. *Every entry of the resolvent (7) has the exact combinatorial expression*

$$(M(t)^{-1})_{ij} = \frac{N_{ij}(t)}{D(t)}, \quad (10)$$

where

$$D(t) = \sum_{\substack{F \subseteq \Omega \\ |F|=n}} w(F) \det(A_F)^2, \quad (11)$$

$$N_{ij}(t) = \sum_{\substack{F \subseteq \Omega \\ |F|=n-1}} w(F) \Delta_i(F) \Delta_j(F). \quad (12)$$

In particular, the coefficient of t^k in either polynomial is obtained by restricting its sum to subsets containing exactly k members of E .

Proof. Let X have the columns $\sqrt{w_q} a_q$ for $q \in \Omega$. Equations (7) and (8) give $M(t) = XX^\top$. Cauchy–Binet applied to this product is exactly (11).

For the numerator, the cofactor formula and Cauchy–Binet on the $(n - 1) \times (n - 1)$ complementary minors give

$$\det M(t) (M(t)^{-1})_{ij} = (-1)^{i+j} \sum_{|F|=n-1} w(F) \det(\widehat{A_F^j}) \det(\widehat{A_F^i}), \quad (13)$$

where a hat means that row is deleted. Expansion of (9) along its first column yields $\Delta_i(F) = (-1)^{i+1} \det(\widehat{A_F^i})$. Substitution in (13) proves (12). Finally, only graph-edge weights contain t , so grouping terms by $|F \cap E|$ gives the coefficient assertion. $\square$

Rooted-forest interpretation

The determinants in (11)–(12) are already finite combinatorial incidence minors. They also have a standard graph description. If $\star \notin F$, put $R = F \cap V$ and $H = (V, F \cap E)$. Then $\det(A_F) \neq 0$ exactly when every component is one of:

- a tree containing exactly one root from R , contributing magnitude 1; or
- an odd-unicyclic component containing no root, contributing magnitude 2.

Consequently $\det(A_F)^2 = 4^u$, where u is the number of odd-unicyclic components. For a nonzero term with $\star \in F$, exactly one component is instead an unrooted tree T_0 ; if its bipartition has sizes p and q , its additional squared factor is $(p - q)^2$. The other components have the two types above.

For completeness, this follows by splitting the incidence matrix into connected blocks. A rooted tree block has determinant magnitude 1; an unsigned unicyclic incidence block is nonsingular precisely for an odd cycle, when its determinant has magnitude 2. For the unique unrooted tree, the signed bipartition vector spans the left nullspace of its incidence matrix, so adjoining $\mathbf{1}$ gives determinant magnitude $|p - q|$. The numerator minors have the same description after adjoining an implicit root at i ; the product in (12) retains the signs required off the diagonal. Under the source’s self-loop convention, the formula itself extends verbatim by using the incidence column $\sqrt{2}e_v$.

Verification, scope, and novelty

The accompanying script exhausts all 74 labelled simple graphs on two, three, or four vertices, at three positive t values per graph. It checks (11)–(12) against direct determinants and inverses in all 222 cases, as well as the exact 3×3 counterexample. This is a sign/index audit, not part of the proof.

A bounded search through 17 August 2026 used the source id/title, citation trails, and close variants of “Conjecture 8.1,” “signless Laplacian inverse entries,” and “exact combinatorial formula $(S + tP)^{-1}$.” It found TU-subgraph formulas for signless-Laplacian characteristic polynomials and specialized inverse formulas, but no answer to Conjecture 8.1 or Problem 8.2. Because Theorem 2 is an elementary all-minors Cauchy–Binet identity, its novelty may be folklore; the claim is only “apparently new within the bounded search.”

The source’s final determinant conjecture is not new: Minghua Lin proved it in arXiv:1212.1934 [2], and the proof appears in the authors’ later Hadamard-type paper [3]. No originality is claimed for that status identification.

Human-review recommendation

First verify the source’s entrywise convention and distinguish the printed conjecture from its stronger motivating repair. Then audit the deleted-row order in (13); once the cofactor signs match (9), the resolvent formula is a direct Cauchy–Binet identity.

References

- [1] C. Hillar, S. Lin, and A. Wibisono, “Inverses of symmetric, diagonally dominant positive matrices and applications,” arXiv:1203.6812 (2012).
- [2] M. Lin, “An upper bound for the determinant of a diagonally balanced symmetric matrix,” arXiv:1212.1934 (2012).
- [3] C. Hillar and A. Wibisono, “A Hadamard-type lower bound for symmetric diagonally dominant positive matrices,” Linear Algebra Appl. 472 (2015), 135–141.